Where Do the Gains Come From? An Audit of Dendritic Context-Gated Protection for Continual Learning

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Abstract—Cortical pyramidal neurons integrate contextual signals in dendritic compartments, and networks with active dendrites inherit this context selectivity. In preliminary, unpublished experiments we proposed DeLP (Dendritic Localized Protection), which freezes dendritic segments for non-current contexts and weights an elastic penalty by each neuron's measured usage under previous tasks; on Permuted-MNIST it appeared to dominate every baseline in both forgetting and accuracy. This paper is the audit we should have run before publication. Prompted by an anomaly — a baseline whose penalty was insensitive to a 100-fold change of its only hyperparameter — we found two defects in our MAS implementation, three deviations from the SI reference, a fixed-fraction pruning schedule that exhausted PackNet's capacity, a missing plain-EWC arm, hyperparameters selected on the test set, and λ grids that left every baseline in its protection-free regime. After correction, DeLP keeps the best five-task accuracy (80.37%, next best 75.27, $p = 0.0024$) but no longer leads on forgetting, and at ten tasks corrected EWC and MAS exceed its accuracy. A seven-rung mechanism ladder localizes what remains: with baselines given the same hidden activation, the entire dendritic chain adds +0.37 points at five tasks ($p = 0.49$); the one mechanism that survives a parameter-budget cap is usage weighting (+1.79, $p = 0.005$ at twenty tasks), worth +4 to +6 points once both sides are gain-calibrated. A hidden-layer gain scalar alone moves accuracy by 16 points — more than the entire method — so the original comparison measured an uncalibrated gain, not a mechanism. Finally, preregistered experiments in a rule-switching task matched to the neuroscience show that compartmentalized routing itself confers no adaptation advantage larger than 30% — the sensitivity limit of 8 preregistered seeds — across three cue-conflict intensities, up to and including fully adversarial cues. We report the corrected accounting as the contribution: when benchmarking protection methods, the gain must be declared, the grid must bracket the optimum, and the baseline must be implemented as carefully as the method.

Index Terms—dendritic computation, continual learning, catastrophic forgetting, active dendrites, elastic weight consolidation, reproducibility

I. Introduction

Cortical pyramidal neurons are not point units: their dendritic arbors form semi-independent compartments that integrate contextual signals with sensory input, and recent in vivo work shows that apical dendrites encode abstract rules which the soma combines with sensory evidence [1]. Networks with active dendrites [2] inherit this context selectivity, and it is tempting to read the same structure as a recipe for continual learning: if dendritic gating is the biological solution to interference within a cell, the structure should indicate where an elastic penalty [3], [4] should protect, and how much [5].

We pursued this idea in preliminary, unpublished experiments (archived; Appendix A) and obtained striking support: a usage-localized elastic penalty (DeLP) appeared to reduce forgetting from -12.67 to -1.94 BWT and raise accuracy from 74.96% to 80.37%, beating Synaptic Intelligence (SI), MAS and PackNet on both axes at five tasks. That comparison, however, carried defects we had not looked for — one instance of a pattern repeated across machine-learning subfields, where re-tuned baselines and honest hyperparameter budgets erase apparent advances [6], [7], [8]. This paper reports the audit, the corrected results, and two lines of evidence that were only possible after it.

How the audit started. In our screening data, MAS was insensitive to a 100-fold change of its only hyperparameter ($\lambda \in \{0.1, 1, 10\}$ moved its BWT by 0.26 points, versus 8.35 for SI and 8.50 for DeLP on the same protocol). A penalty that does not respond to its own strength is not being applied. Chasing that one observation exposed two MAS bugs, three deviations from the SI reference, a capacity-exhausting PackNet schedule, a missing EWC arm, and two protocol errors (test-set selection and degenerate λ grids) that systematically favored the proposed method.

What survives. We re-ran the full comparison with every arm implemented and selected correctly (Section IV); decomposed the method into seven mechanisms on one pipeline (Section V); and ran three preregistered experiments in a rule-switching task matched to the neuroscience, asking whether compartmentalized routing itself helps adaptation (Section VI). Three conclusions survive, all smaller than the original claims: usage weighting is the one dendritic mechanism with evidence that survives budget capping and gain calibration; the dominant factor in the original table was an uncalibrated hidden-layer gain;

and compartmental structure, on its own, buys nothing measurable in either paradigm.

Contributions.

1) A defect-by-defect audit of our own preliminary comparison, with corrected eight-arm results at five and ten tasks and paired tests (Section IV).
2) A mechanism ladder isolating seven candidate sources of gain, plus a gain-sweep analysis showing that a single scalar — hidden-layer gain — outweighs the entire method, with a probe showing the proposed gate never opens (Section V).
3) A calibration protocol separating "default", "capped-budget" and "gain-calibrated" comparisons, under which usage weighting is the only mechanism that survives (Section V-A).
4) Preregistered rule-switching experiments (with instrument-incident gates) showing no adaptation benefit of compartmentalized routing across three cue-conflict intensities, with a post hoc sensitivity analysis bounding the claim to effects above a 30% adaptation reduction (Section VI).

II. Background and Related Work

Dendritic computation. Cable theory and two-layer models made the theoretical case for dendritic nonlinearity [9], and its computational vocabulary is mapped in detail [10], [11]; one cortical neuron can approximate a deep network [12], and apical dendrites were proposed as excitable integrators long ago [13], [14]. Dendritic-plasticity work argues that context-dependent inhibition gates plasticity at branches, letting overlapping assemblies coexist [15]. On the engineering side, dendritic complexity is argued to be a route to neuromorphic efficiency [16], [17], [18], [19].

Active dendrites and continual learning. Neurons with multiple dendritic segments that receive a context vector and gate the feedforward response support multi-task learning [2]. Regularization methods protect important weights (EWC's Fisher [3], SI's path integral [4], MAS's output-norm sensitivity [20]); isolation methods allocate capacity (PackNet [5], hard attention [21]); context-dependent gating modulates units by context [22]. DeLP sits between the families: a Fisher-style penalty whose per-parameter weight comes from the gating structure itself [23]. The audit protocol used here — validation-selected grids, parameter-budget caps, gain calibration, preregistered gates — follows the reporting standards advocated for empirical ML [24], [25].

III. Method

Active-dendrite neuron. Each hidden neuron j ($j = 1..N$) carries K dendritic segments. A context vector $\mathbf{c} \in \mathbb{R}^T$ (one-hot task identity) is projected onto segments through $U \in \mathbb{R}^{K\times N\times T}$, the dendritic response is the maximum over segments, and

$$h_j = \mathrm{kWTA}\big((\mathbf{w}_j^\top \mathbf{x} + b_j)\cdot \sigma(d_j)\big), \tag{1}$$

with σ the logistic function and kWTA keeping the top-k activations per layer ($k = N/4$). During training of task t, gradients to U are masked so only the rows selected by context t are updated (segment freeze).

Why the penalty looks at the shared projection. An interference-channel ablation established where forgetting lives: freezing dendritic segments changes little (-14.79 vs. -15.16 BWT for unfrozen); extending the freeze to the feedforward rows of task-owned neurons collapses accuracy to 67.35% (capacity starved); per-task readouts relocate the interference into the shared feedforward projection (-19.78 BWT). The shared feedforward projection and readout are therefore the channels a protection method must address, and the dendritic gate supplies a per-neuron usage statistic for grading the protection: $o_{t',j}$, the fraction of task-t' samples on which neuron j is inside the kWTA winner set. The penalty is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \sum_{t'<t} \sum_i F^{\mathrm{loc}}_{t',i}\,(\theta_i - \theta^*_{t',i})^2, \quad F^{\mathrm{loc}}_{t',i} = o_{t',j(i)} \cdot F_{t',i}, \tag{2}$$

with $F_{t'}$ the diagonal Fisher estimated on 300 samples per task and o on 500; both reuse the task's training data. Unlike hard gating, protection is soft: a penalized neuron can still move at a quadratic price. Fisher uses the max-log-probability variant throughout (Section VII). Test-time context inference from a 500-sample buffer (linear classifier on feedforward features) matched oracle context exactly.

IV. Auditing the original comparison

The original comparison (five seeds, paired t-tests, one pipeline) reported DeLP at -1.94 ± 0.42 BWT / 80.37 ± 0.67% accuracy, ahead of SI (-3.39 / 77.56), MAS (-12.32 / 75.27) and PackNet (0.00 / 43.95) — these are the archived original numbers (exp12_result.json); the SI rows of Table I report the corrected re-run under the validation-selected protocol and therefore differ. The audit found seven defects; we state each with its location and consequence.

D1: MAS importance estimated from batch-averaged gradients, twice divided. Our MAS backward-passed an already-averaged objective, multiplied squared mean gradients by $1/B$ again (losing sample variance and shrinking Ω by the batch size), and then divided the accumulated Ω by the usage count at the end of every task without resetting, so task 0's contribution decayed as $(1/8)^t$ (7.5×10^{-9} by task 9). The penalty protected only the most recent task. Consequence: MAS's λ elasticity was $+0.08$ over two orders of magnitude; after the fix it is $+1.40$, and its best five-task BWT improves from -12.07 to -3.60. The original "MAS trails by 10.4 points" was mostly this bug.

D2: SI deviated from its reference in three places. The denominator used $|\Delta\theta|$ instead of $(\Delta\theta)^2 + \xi^2$ (with $\xi = 10^{-3}$ instead of 0.1), and an abs() destroyed the sign of the path integral. The effect on BWT at matched optimum is

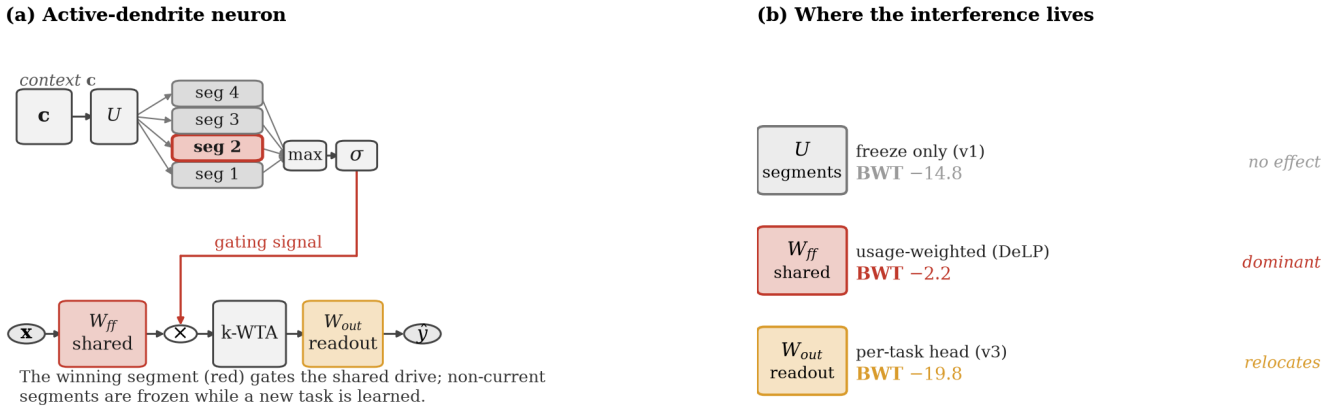


Fig. 1. (a) The active-dendrite neuron: a context vector is projected onto K dendritic segments, the strongest segment gates the shared feedforward drive, and a kWTA nonlinearity sparsifies the layer; only the winning segment is updated during a task. (b) The three candidate interference channels and their measured consequences at five tasks: freezing the dendritic segments changes little; per-task readouts relocate the interference; hard per-neuron freezing collapses capacity.

small (-0.15 vs. -0.36), but the implementation was not SI.

D3: PackNet pruned 20% of free capacity per task. After five tasks 32.8% of parameters remain, after ten 10.7%: the reported collapses to 43.95% / 27.38% are budget exhaustion, not method failure. The corrected variant plans per-task capacity proportionally and prunes per layer.

D4: no plain-EWC arm. DeLP is an EWC penalty plus branch protection, so without a plain-EWC arm the decomposition "how much is EWC, how much is the dendritic part" was unanswerable. Added.

D5: hyperparameters selected on the test set. Screening evaluated λ candidates on the test set. Corrected protocol: held-out validation split (train[10000:12000]). The validation split is applied to the three arms whose re-implementation evaluates it (plain EWC, corrected SI, corrected MAS). The arms that reuse the original implementation keep their original λ, because those implementations do not record held-out metrics; we state this asymmetry rather than claim a uniform rule, and it applies equally to the proposed arm and to the baselines.

D6: degenerate λ grids. The best λ of DeLP, SI and MAS all sat on the upper bound of their grids: the baselines were left in their protection-free regime while the proposed method's grid covered its working range. Extending SI's grid to $\lambda = 100$ moves its BWT from -4.04 to -0.36. A related selection rule — choosing λ by best BWT — is structurally degenerate here, since BWT improves monotonically with λ while accuracy is unimodal: across 21 (mechanism $\times$ task-count) cells, BWT-based selection hit the grid upper bound 21 times, each 8–31 accuracy points below the cell's peak. All corrected results select λ by validation accuracy.

D7: initialization not seeded (protocol). An earlier unseeded pipeline drifted in the third digit and once flipped a verdict (SI advantage $p = 0.058$ unseeded vs. $p = 0.039$ seeded). All corrected runs seed every arm.

Withdrawn suspicion. Architectural parameter inequality between the proposed arm and the baselines (+1.25%) was re-measured and is negligible; we withdraw it as a concern.

Corrected comparison. Table I replaces the original main table. Three original claims do not survive. (i) "MAS trails by 10 points": the corrected MAS trails by 1.8 accuracy-significant points at five tasks ($p = 0.015$ on BWT, $p = 0.0001$ on accuracy) and is statistically indistinguishable on BWT at ten tasks (-5.22 vs. -5.38, $p = 0.63$) with 7.7 points more accuracy ($p = 0.0001$). (ii) "DeLP forgets less than SI": at ten tasks SI's BWT is significantly better (-1.31 vs. -5.38, $p < 0.0001$). (iii) "best accuracy at ten tasks": corrected EWC (68.35%, $p = 0.0003$) and MAS (62.91%, $p = 0.0001$) both exceed DeLP's 55.23%. What survives: the best five-task accuracy, 5.1 points above the next best arm ($p = 0.0024$); the orig. rows show the original table compared against broken baselines. The table is best read as a frontier, not a ranking: SI occupies the low-forgetting end, corrected EWC/MAS the high-accuracy end at ten tasks, and DeLP the best measured five-task operating point.

Second benchmark. Does the corrected picture transfer? Re-running the full corrected protocol (identical arms, grids, val λ rule and seeds) on Permuted-Fashion-MNIST gives the same shape (Table II): DeLP holds the best five-task accuracy (+6.2 over corrected MAS, $p = 0.0003$), the ten-task accuracy lead disappears (MAS +0.8, $p = 0.54$; EWC $p = 0.32$), and corrected SI/MAS forget significantly less at ten tasks ($p \leq 0.003$). The conclusion of the audit is not an artifact of one dataset.

V. Mechanism ladder and the gain confound

The corrected table says the comparison was flawed, not which part of the method works. To attribute gains to mechanisms we trained seven rungs on one pipeline,

TABLE I

Corrected comparison, five seeds, paired t-tests against DeLP. The corrected arms (plain EWC, corrected SI, corrected MAS) use their own best λ selected on the held-out validation split (MNIST train[10000:12000]); the remaining arms keep the λ reported by the original run, because their original implementations do not record validation metrics — this is stated explicitly rather than glossed over, and it is the same asymmetry the original table had. Grids per family are released with the code and verified to bracket each optimum (grid_check.md). orig. rows reproduce the original implementations (verified bit-identical on shared seeds). BWT: higher is better; acc: higher is better.

	Five tasks		Ten tasks	
Method	BWT	Final avg acc	BWT	Final avg acc
DeLP (ours)	$\mathbf{-1.94 \pm 0.42}$	$\mathbf{80.37 \pm 0.67}$	-5.38 ± 0.44	55.23 ± 1.41
EWC (online, multi-anchor)	-7.08 ± 1.19	74.28 ± 0.86	-7.81 ± 1.28	$\mathbf{68.35 \pm 0.95}$
SI (reference impl.)	$\mathbf{-0.03 \pm 0.35}$	62.33 ± 1.41	$\mathbf{-1.31 \pm 0.29}$	40.57 ± 1.51
SI (orig. impl.)	-0.15 ± 0.38	63.34 ± 1.63	-0.86 ± 0.13	39.46 ± 1.46
MAS (corrected)	-3.72 ± 0.82	75.22 ± 0.55	-5.22 ± 0.66	62.91 ± 1.02
MAS (orig. impl.)	-12.32 ± 2.53	75.27 ± 1.99	-19.88 ± 1.27	68.00 ± 1.23
PackNet (corrected)	-12.67 ± 2.60	74.96 ± 2.01	-19.92 ± 0.68	67.93 ± 0.83
PackNet (orig. impl.)	0.00 (mask)	43.95 ± 0.82	0.00 (mask)	27.38 ± 0.63

TABLE II

Corrected comparison on Permuted-Fashion-MNIST (second benchmark; 5 seeds, val-selected λ, same protocol as Table I). Original-implementation rows, omitted here, again trail their corrected counterparts by 5–11 accuracy points.

	Five tasks		Ten tasks	
Arm	acc.	BWT	acc.	BWT
DeLP	66.9±0.9	−8.5±1.4	55.5±1.6	−18.0±1.6
MAS (corrected)	60.8±0.7	−4.1±1.5	56.3±1.1	−8.5±1.3
EWC	57.1±1.8	−11.3±1.2	53.5±3.4	−12.7±3.0
PackNet (corr.)	55.3±3.1	−15.2±3.2	48.4±3.9	−22.2±4.3
SI (corrected)	34.6±2.7	**−2.8**±0.9	26.6±1.2	**−1.9**±0.6

TABLE III

Rung-by-rung accuracy gain (each rung at its own validation-selected λ; p in parentheses; five seeds). Differences are computed from unrounded per-seed means, so subtracting the rounded rung accuracies in the released table can differ by 0.01.

Step	T=5	T=10	T=20
R0→R0b: remove squashing	+5.37 (.000)	+4.49 (.000)	+2.13 (.012)
R0b→R1: dendritic params	+0.00 (1.00)	+0.00 (1.00)	+0.00 (1.00)
R1→R2: context gating	+0.61 (.172)	+0.37 (.019)	+1.35 (.260)
R2→R3: segment freeze	−0.11 (.616)	+0.05 (.873)	−0.37 (.602)
R3→R4: usage weighting	−0.13 (.563)	+0.07 (.820)	+1.02 (.008)
R4→R5: CE Fisher	+0.04 (.842)	+0.04 (.863)	−0.76 (.404)
R4 (full method) vs. R0b	+0.37 (.494)	+0.49 (.083)	+2.00 (.025)

each rung adding exactly one mechanism, all else identical (R0: point MLP + EWC with squashed hidden activation; R0b: value-range compression removed; R1: + dendritic parameters, context pinned to zero; R2: + context gating; R3: + segment freeze; R4: + usage weighting (= DeLP); R5: + CE Fisher). Two self-checks gate the analysis: R1 must be bit-identical to R0b (it is: $\Delta = 0.00$, $p = 1.000$, confirming the padded dendritic layer is dead weight), and every rung's λ elasticity must be positive (all are, +2.55 to +7.94 in the released ladder batch).

Table III is the paper's central accounting. The single largest step is the removal of the hidden squashing: sigmoid compresses the pre-activation into $(0, 1)$, collapsing amplitude information (a gain sweep shows accuracy rises monotonically with the post-sigmoid gain, $r = +0.94$ — amplitude compression, not "saturation", is the mechanism). The dendritic parameters themselves contribute exactly zero (bitwise identity). Context gating is small, positive and direction-stable (BWT +3.10, $p = 0.001$ at five tasks). Segment freeze is zero. Usage weighting is the only rung whose value grows with sequence length (+1.02, $p = 0.008$ at twenty tasks), as expected if it offsets penalty amortization across more anchors. Summed against the same-activation baseline R0b, the full method is +0.37 ($p = 0.49$) at five tasks, +0.49 ($p = 0.083$) at ten, and +2.00 ($p = 0.025$) at twenty.

The gain confound, quantified. kWTA is scale-equivariant ($\mathrm{kWTA}(\alpha t) = \alpha\,\mathrm{kWTA}(t)$, verified exactly), so a hidden-layer gain α does not change the function class — but it changes how far weights must travel, and EWC-style penalties are quadratic in displacement. Sweeping α on plain "MLP+EWC" moves five-task accuracy from 76.24 ($\alpha_{\text{eff}} \approx 0.5$) to 92.40 ($\alpha$=16) — +16.2 points, versus +0.37 for the entire method chain; at α=2 already, plain MLP+EWC beats the full DeLP on accuracy and forgetting at all three task counts. The DeLP gate is $t \cdot \sigma(d)$ with d learnable, so it could in principle open; a probe that logs the gate during full training shows it never does: $\sigma(d) = 0.575$–0.577 across all five tasks (0.55 at twenty), i.e., an uncalibrated ≈ 0.58 gain sitting near the worst end of the sweep. The original comparison therefore measured a gain miscalibration, not a mechanism.

A. What survives: calibration protocols

Because gain interacts with every λ-style penalty, "how much better is the method" depends on the calibration protocol, and we now report all three (Table IV). Under default settings (neither side gain-tuned; dendritic parameter budget capped at $nseg = \min(8, T)$, which cuts the twenty-task dendritic block from 102,400 to 40,960 parameters with all five rungs unchanged within noise), usage

TABLE IV
The same question under three calibration protocols: DeLP accuracy advantage over the best same-activation baseline.

Protocol	Adv.	Status
Both default (published cfg.)	−0.85 to +3.7	changes sign with T; unusable
Usage weighting, capped budget	+1.79 (p=.005)	T=20 only; budget-controlled
Both gain-calibrated	+4 to +6	two-point probe; lower bound

weighting beats the same-activation baseline by +1.79 ($p = 0.005$) at twenty tasks. Under gain calibration (two-point gain probe per family at each family's own optimal λ), DeLP gains +6.3/+8.6/+9.7 points (five/ten/twenty tasks) — but so do the baselines (MAS +4.1/+3.4/+4.6, SI +3.3/+6.0/+3.1, EWC +3.5/+0.7/+3.0); the calibrated comparison leaves DeLP ahead by +4 to +6 points (87.81/83.00/75.18 vs. best baseline 81.80/78.99/70.92). Under the original protocol (neither side tuned, published configurations) the difference changes sign with T (−0.85 to +3.7) and is unusable. The usage-weighting result is a per-family two-point probe at fixed λ^*, not a joint (gain $\times$ λ) grid, so +4–6 is an approximate lower bound.

VI. Preregistered rule-switching tests

The biological motivation predicts more than a usage statistic: if apical compartments are where top-down rule signals meet sensory evidence [1], a network that routes the two into separate compartments should adapt faster when rules change than a network that fuses them. This is a structural claim, distinct from protection, and we tested it in a preregistered companion study (thresholds, gates and sample sizes fixed before running; deviations require a numbered amendment).

Task and models. Rule-switching categorization: $d = 16$ stimuli, 4 random linear rules (resampled per seed), unsignaled switches every 80 trials, 480-trial episodes, reward correctness $p = 0.85$. GRU-backed policy networks trained with REINFORCE (1500 episodes, 8 seeds, lr 10^{-3}, belief-auxiliary loss). DC-Net routes a rule-cue plus history summary and reward prediction error to a top (apical) compartment feeding a gating signal, while sensory input reaches the basal dendrites; Point-Net receives the identical information fused into one stream; Random-MLP has the same parameters with scrambled structure; a trivial-learner gates task difficulty. In PR-002 each block carries a one-hot rule cue, valid with $p = 0.9$ and actively misleading otherwise, giving blocks where the top-down signal conflicts with sensory evidence — the regime where compartmental isolation should help most. PR-003 removes the contrast entirely: the cue is misleading in every block ($p_{\text{valid}} = 0$), so the top-down stream carries pure inverted information and must be suppressed rather than occasionally corrected.

Result. All three hypotheses fail to meet the preregistered criterion (Table V). In PR-001, with validity gates passed

TABLE V
Preregistered rule-switching results across three cue-conflict intensities (cue misleading in 0%, 10%, or 100% of blocks; per-seed medians of post-switch trials to 80% correct, 8 seeds; Mann-Whitney U, one-sided; reduction threshold ≥25%).

	PR-001 (none)	PR-002 (10%)	PR-003 (100%)
DC-Net final acc	0.648	0.649	0.647
Point-Net final acc	0.643	0.654	0.632
Trivial learner acc	0.500	0.500	0.500
DC-Net adapt med. [CI]	6.5 [4.5, 7.8]	6.5 [3.9, 7.6]	6.5 [4.5, 7.8]
Point-Net adapt med.	7.0	7.0	6.5
Median reduction	7.1%	7.1%	0.0%
p	0.335	0.296	0.458
Cohen's d	−0.27	−0.18	−0.37

(trivial learner 14.8 points below; scrambled-structure control indistinguishable from DC-Net, so the null cannot be attributed to parameter count), DC-Net's post-switch adaptation is 7.1% faster at $p = 0.335$. In PR-002, with the misalignment rate verified at 9.83% ($n = 240{,}000$ switches) against the preregistered 10%, the numbers are essentially identical (7.1%, $p = 0.296$); in cue-valid blocks the two architectures are exactly tied (6.0 vs. 6.0, $p = 0.69$). In PR-003, with misalignment verified at 100%, the two are again exactly tied (6.5 vs. 6.5, $p = 0.458$) — and both medians sit at this task's measurement floor, where detection is hardest. Validity gates passed in all three runs; a first instrumented run of PR-002 was automatically voided by its own preregistered gate when a logging defect left condition labels unrecorded, and re-run — we report this because the gate doing its job is part of the result.

Sensitivity. The claims above are bounded, and we bound them ourselves. A post hoc simulation on the Point-Net reference distribution (20,000 resamples per point; DC-Net samples drawn as a fixed scaling of bootstrap draws) shows that with 8 seeds and a one-sided Mann-Whitney test at $\alpha = 0.05$, the design had 80% power only against median reductions of ≥30%; at the preregistered 25% threshold power was ≈50%, and at the observed 7.1% it was <5%. The defensible conclusion is therefore: no compartmentalization advantage of 30% or more exists in this task family, while smaller effects — including the observed 7.1% — cannot be distinguished from zero with this sample size. In PR-003 both medians sit at the measurement floor, degrading the detectable effect to 50%; an exact tie at the floor nonetheless weakly favors the null.

Reading. In this task family a GRU history summary already carries the rule inference; routing top-down information through a separate compartment adds nothing measurable, at any of three conflict intensities — none, occasional, or fully adversarial — and the conflict gradient itself shows no dose–response trend (adaptation may sit near the information-theoretic floor for 4-way rules under 15% feedback noise). Combined with the ladder: the measurable value of dendritic structure in our systems is not its routing or its isolation but the measurement it computes — per-neuron, per-context usage — and even that value appears only as a weighting inside an elastic

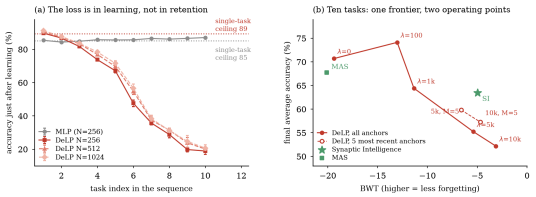


Fig. 2. Ten-task learning/retention decomposition: the loss lies in the learning term.

penalty, only under a capped budget, and only at long horizons.

VII. Learning versus retention at ten tasks

For completeness we retain the decomposition $\bar{a}_T = \ell + \frac{T-1}{T}\mathrm{BWT}$ (ℓ: mean just-learned accuracy). At ten tasks the point MLP learns each task to ≈86% and forgets heavily, while DeLP learns less and forgets little; the deficit is not a width limit (256→1024 recovers 3.4 points of ℓ, costs 2.5 of BWT) nor a representational wall (single-task ceiling 89.4% at N=256), but an accumulated elastic anchor cost: sweeping λ trades the two terms monotonically (BWT improves 16.3 points while ℓ falls 33.2 from λ=100 to 10^4), and truncating the anchor set to the five most recent tasks recovers part of ℓ at every λ. Per-task kWTA winner sets overlap substantially (mean pairwise Jaccard 0.51): gating does not partition neurons — "ownership" of parameters is imposed, not discovered. No single λ dominates SI on both axes at ten tasks (Fig. 2).

VIII. Discussion

What the dendritic story buys — revised. The audited evidence supports a claim with three qualifiers. Dendritic compartmentalization provides no capacity (bitwise-null when context is pinned), no protection by itself (segment freeze ≈ 0), and no adaptation advantage above the 30% sensitivity limit in preregistered rule-switching tasks across three conflict intensities, including fully adversarial top-down signals. What it provides, once, and only as a component: context-conditioned gating computes a per-neuron usage statistic, and weighting an elastic penalty by that statistic is the one mechanism whose gain survives budget capping (+1.79, $p = 0.005$ at twenty tasks) and grows under calibration. This is closer to an accounting principle than to an architecture: the biology says where to look, standard elastic machinery does the work, and every other presumed benefit dissolved under controlled comparison.

Lessons for benchmarking protection methods. Five practices would have caught most of Section IV before publication: (i) a hyperparameter-elasticity check per arm (a penalty that ignores its own λ is not applied); (ii) validation-based model selection with grids verified to bracket each arm's optimum on both ends; (iii) a plain-EWC arm whenever the method contains EWC as a component; (iv) an activation- and gain-matched baseline, since quadratic penalties interact with parameter scale and a single free scalar outweighed the entire method here; (v) preregistered gates for confirmatory claims, including gates that void the run when instrumentation fails.

Limitations. (i) Both benchmarks are MNIST-derived (Permuted-MNIST, Permuted-Fashion-MNIST); a non-MNIST family such as Split-CIFAR10 remains future work. (ii) Task identity is available at training time (test-time inference shown lossless here). (iii) Gain calibration uses a two-point probe per family at fixed λ^*, not a joint (gain × λ) grid: the +4–6 calibrated advantage is an approximate lower bound. (iv) The one-hot context leaves the capped dendritic block +20% parameters at twenty tasks. (v) The rule-switching null covers one task family (4-way rules, 85% feedback noise, GRU summary); gates cannot rule out effects below ∼7% at this sample size. (vi) All runs use 2 epochs and 10k samples per task.

IX. Conclusion

We asked whether dendritic context gating tells a continual-learning system where to protect its knowledge, and the audited answer is more modest and more useful than our first report. The original comparison was inflated by baseline defects, test-set selection and an uncalibrated gain; after correction, the dendritic structure contributes exactly one mechanism with standing evidence — a per-neuron usage statistic that localizes an elastic penalty (+1.79 points at twenty tasks under a capped budget, +4–6 gain-calibrated) — and nothing else: no capacity, no protection from freezing, no adaptation advantage from compartmental routing across three preregistered conflict levels. A hidden-layer gain scalar moved accuracy more than the entire method — both the reason the original table looked as it did and the paper's most transferable finding. Dendritic computation, in this benchmark family, is most useful not as extra capacity or isolation but as measurement: a signal about which parts of a network mean what — provided the comparison is calibrated well enough to see it.

Appendix A
Reproducibility and hyperparameters

Code and raw result files (preregistrations, per-seed JSONs, analysis scripts) are archived at https://github.com/zhouranxi/delp-audit. Original experiments: dendritic-lab/ — exp4*/exp5 (ablation chain, DeLP), exp6 (original baselines), exp8 (shared pool, context inference), exp9 (single-pipeline table), exp10 (decomposition), exp11 (frontier), exp12 (seeded re-run). Audit and re-analysis: bench/ — exp6_fixed.py (corrected baselines, validation λ, EWC arm), ladder.py (945 cells, zero failures), kwta_scale_check.py, gate_probe.py, sig_probe.py, analyze_ei.py. Rule-switching study: dendrite-mve/ with preregistrations prereg_pr00{1,2,3}_*.md (amendments, corrections, result records) and raw per-seed JSONs (results/, results_cue/, results_cue_adv/). Every run

torch.manual_seeds before model construction; settings not stated in the text: $N = 256$, $K = T$ segments (8, top-3, in the capped variant), kWTA $k = N/4$, Fisher/usage from 300/500 samples, five seeds (eight for the rule-switching study), one RTX 3090. Corrected-baseline runs total 80 cells; ladder and calibration runs 665 cells.

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